```sql
USE PRACTICE1;

---Retrieving all data in malaria_dataset from the database
SELECT * from malaria_data1;

--Counting(Confirming) the Available Columns
SELECT COUNT(DISTINCT(column1)) FROM malaria_data;




--Counting the total Number of Columns Present
SELECT COUNT(*) AS NumberOfColumns
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'malaria_data1'



--Renaming Columns

EXEC sp_rename 'malaria_data.[column1]',  'Region', 'COLUMN';
EXEC sp_rename 'malaria_data.[column2]',  'District', 'COLUMN';

EXEC sp_rename 'malaria_data1.[0-28_DAYS_Female]',  '0_28Days_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_0_28_DAYS_MALE]',  '0_28Days_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_0_28_DAYS_TOTAL]',  '0_28Days_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_10_19_YEARS_FEMALE]',  '10_19Days_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_10_19_YEARS_MALE]',  '10_19Days_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_10_19YRS_TOTAL]',  '10_19Days_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_20_YRS_FEMALE]',  '20+YRS_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_20_YRS_MALE]', '20+YRS_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_20_YRS_TOTAL]', '20+YRS_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_5_9YRS_FEMALE]', '5_9YRS_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5_9YRS_MALE]', '5_9YRS_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5_9YRS_TOTAL]', '5_9YRS_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_29_DAYS_4YRS_FEMALE]', '29Days_4Yrs_Female', 'COLUMN';
```

```sql
EXEC sp_rename 'malaria_data1.[_29DAYS_4YRS_MALE]', '29Days_4Yrs_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_29_DAYS_4YRS_TOTAL]', '29Days_4Yrs_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_]', '0_28Days(BSRDT)_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_0_28_DAYS_B_S_RDT_MALE]', '0_28Days(BSRDT)_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_0_28_DAYS_B_S_RDT_TOTAL]', '0_28Days(BSRDT)_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_10_19_YRS_B_S_RDT_FEMALE]', '10_19YRS(BSRDT)_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_10_19_YRS_B_S_RDT_MALE]', '10_19YRS(BSRDT)_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_10_19_YRS_B_S_RDT_TOTAL]', '10_19YRS(BSRDT)_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_20_B_S_RDT_FEMALE]', '20+YRS_(BSRDT)_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_20_B_S_RDT_MALE]', '20+YRS_(BSRDT)_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_20_B_S_RDT_TOTAL]', '20+YRS_(BSRDT)_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_29DAYS_4YRS_B_S_RDT_FEMALE]', '29Days_4Yrs(BSRDT)_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_29DAYS_4YRS_B_S_RDT_MALE]', '29Days_4Yrs(BSRDT)_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_29DAYS_4_YRS_B_S_RDT_TOTAL]', '29Days_4Yrs(BSRDT)_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_5_9YRS_B_S_RDT_FEMALE]', '5_9YRS(BSRDT)_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5_9YRS_B_S_RDT_MALE]', '5_9YRS(BSRDT)_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5_9YRS_B_S_RDT_TOTAL]', '5_9YRS(BSRDT)_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[TOTAL_CASES_CONFIRMED_B_S_RDT]', 'TotalCasesConfirmed_BSRDT', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_0_28_DAYS_FEMALE_CT]', '0_28Days_CT_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_0_28_DAYS_MALE_CT]', '0_28Days_CT_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_0_28_DAYS_TOTAL_CT]', '0_28Days_CT_Total', 'COLUMN';

```

```sql
EXEC sp_rename 'malaria_data1.[_10_19_YRS_FEMALE_CT]', '10_19YRS_CT_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_10_19_YRS_MALE_CT]', '10_19YRS_CT_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_10_19_YRS_CT_TOTAL]', '10_19YRS_CT_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_20_YRS_FEMALE_CT]', '20+YRS_CT_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_20_YRS_MALE_CT]', '20+YRS_CT_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_20_YRS_CT_TOTAL]', '20+YRS_CT_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_29DAYS_4YRS_FEMALE_CT]', '29Days_4Yrs_CT_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_29_DAYS_4YRS_MALE_CT]', '29Days_4Yrs_CT_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_29_DAYS_4YRS_CT_TOTAL]', '29Days_4Yrs_CT_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_5_9YRS_FEMALE_CT]', '5_9YRS_CT_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5_9YRS_MALE_CT]', '5_9YRS_CT_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5_9YRS_CT_TOTAL]', '5_9YRS_CT_Total', 'COLUMN';
EXEC sp_rename 'malaria_data1.[PREG_10_19YRS]', 'Preg_10_19', 'COLUMN';
EXEC sp_rename 'malaria_data1.[PREG_20_YRS]', 'Preg_20Plus', 'COLUMN';
EXEC sp_rename 'malaria_data1.[TOTAL_PREG]', 'Preg_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[TOTAL_CASES_FEMALE]', 'TotalCases_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[TOTAL_CASES_MALE]', 'TotalCases_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[GRAND_TOTAL_CASES]', 'GrandTotalCases', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_5YRS_FEMALES_TOTAL]', 'Under5_Female_Total', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5YRS_MALE_TOTAL]', 'Under5_Male_Total', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5YRS_GRAND_TOTAL]', 'Under5_GrandTotal', 'COLUMN';

EXEC sp_rename 'malaria_data1.[AREA_Km2]', 'Area_km2', 'COLUMN';
EXEC sp_rename 'malaria_data1.[TOTAL_POPULATION]', 'TotalPopulation', 'COLUMN';
EXEC sp_rename 'malaria_data1.[POPULATION_DENSITY_People_per_km2]',
```

```sql
    'PopulationDensityPerKm2', 'COLUMN';

EXEC sp_rename 'malaria_data1.[CORDINATES_LATITUDE_N]', 'Latitude', 'COLUMN';
EXEC sp_rename 'malaria_data1.[CORDINATES_LONGITUDE_E]', 'Longitude', 'COLUMN';

EXEC sp_rename 'malaria_data1.[Year]', 'Year', 'COLUMN';
EXEC sp_rename 'malaria_data1.[Incidence_Rate]', 'IncidenceRate', 'COLUMN';
EXEC sp_rename 'malaria_data1.[POPULATION_DENSITY]', 'PopulationDensity', 'COLUMN';


--Number of Rows
SELECT COUNT(*)  From malaria_data;


--19/01/2026
--Changing Datatypes

  ALTER TABLE malaria_data1 ALTER  COLUMN Region NVARCHAR(30);
  ALTER TABLE malaria_data1  ALTER  COLUMN District NVARCHAR(30);
  ALTER TABLE malaria_data1  ALTER COLUMN "0-28Days_Female" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days_Male" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days_Total" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "10_19Days_Female" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "10_19Days_Male" INT;
  ALTER TABLE malaria_data1 ALTER  COLUMN "10_19Days_Total" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_Female" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_Male" INT;
  ALTER TABLE malaria_data1 ALTER  COLUMN "20+YRS_Total" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS_Female" INT;
  ALTER TABLE malaria_data1 ALTER  COLUMN "5_9YRS_Male" INT;
  ALTER TABLE malaria_data1 ALTER  COLUMN "5_9YRS_Total" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_Female" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_Male" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_Total" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days(BSRDT)_Female" INT;
  ALTER  TABLE malaria_data1 ALTER  COLUMN "0_28Days(BSRDT)_Male" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days(BSRDT)_Total" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "10_19YRS(BSRDT)_Female" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "10_19YRS(BSRDT)_Male" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "10_19YRS(BSRDT)_Total" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_(BSRDT)_Female" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_(BSRDT)_Male" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_(BSRDT)_Total" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs(BSRDT)_Female" INT;
  ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs(BSRDT)_Male" INT;
```

```sql
ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs(BSRDT)_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS(BSRDT)_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS(BSRDT)_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS(BSRDT)_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "TotalCasesConfirmed_BSRDT" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days_CT_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days_CT_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "10_19YRS_CT_Female" INT;
ALTER  TABLE malaria_data1 ALTER COLUMN "10_19YRS_CT_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "10_19YRS_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_CT_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_CT_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_CT_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_CT_Male" INT;
ALTER  TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS_CT_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS_CT_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Preg_10_19" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Preg_20Plus" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Preg_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "TotalCases_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "TotalCases_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "GrandTotalCases" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Under5_Female_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Under5_Male_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Under5_GrandTotal" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Area_km2" DECIMAL(18,8);
ALTER TABLE malaria_data1 ALTER COLUMN TotalPopulation DECIMAL(18,8);
ALTER TABLE malaria_data1 ALTER COLUMN "PopulationDensityPerKm2" DECIMAL (18,8);
ALTER TABLE malaria_data1 ALTER COLUMN "Latitude" DECIMAL(9,6);
ALTER TABLE malaria_data1 ALTER COLUMN "Longitude" DECIMAL(9,6);
ALTER TABLE malaria_data1 ALTER COLUMN "Year" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "IncidenceRate" DECIMAL(18,8);
ALTER TABLE malaria_data1 ALTER COLUMN "PopulationDensity" DECIMAL (18,8);

--Checking out New datatypes
EXEC sp_help malaria_data1;

--Realise Duplicates in Region Column and District Column
SELECT DISTINCT(Region)
FROM malaria_data
WHERE Region IN (
      SELECT DISTINCT(District)
      FROM malaria_data
```

```sql
  ) AND  YEAR = '2020'
  ORDER BY Region;


    --Realise Duplicates in Region Column and District Column
  SELECT DISTINCT(TotalCasesConfirmed_BSRDT) , Region
  FROM malaria_data
  WHERE Region IN (
        SELECT DISTINCT(District)
        FROM malaria_data
  ) AND Year = '2020'
  ;



    ---------------------------------------------------------------------
    ---------------------------------------------------------------------
    ---------------------------------
  /** PERFOMING DATA QUALITY CHECKS ON B/S & RDT confirmed **/

  WITH Total_cases As(
  SELECT Region, SUM(TOTALCasesConfirmed_BSRDT) As TotalConfirmed_2020
  FROM malaria_data
  WHERE Region IN (
        SELECT DISTINCT District
        FROM malaria_data
  )
  AND YEAR = '2020'
 GROUP BY Region
  --ORDER BY Region
  )
  --SELECT * FROM Total_caseS;
  SELECT SUM(TotalConfirmed_2020) As Total_cases
  FROM Total_cases;


    ---------------------------------------------------------------------
    ---------------------------------------------------------------------
    ---------------------------

  --Confirming Total Cases Confirmed(B/S & RDT) As per 2020 per Region
  SELECT Region, SUM(TotalCasesConfirmed_BSRDT) AS TotalConfirmed_2020
FROM malaria_data
WHERE Region IN (
    SELECT DISTINCT District
    FROM malaria_data
)
AND Year = 2020
GROUP BY Region
```

```sql
ORDER BY Region;

-------------------------------------------------------------------------------------------------------------------
--Total Cases Confirmed(B/S & RDT) from all 15 Ugandan Regions as per 2020
WITH Total_Cases2_cte AS(
        SELECT
        Region,
        TotalCasesConfirmed_BSRDT
        FROM malaria_data
        WHERE Region = District
        AND Year = 2020
        )
SELECT SUM(TotalCasesConfirmed_BSRDT) As Total_Cases1
FROM Total_Cases2_cte;


-------------------------------------------------------------------------------------------------------------------
--Confirming Total Cases for [10-19Days_Total] registered in  Lango Region as per 2020
WITH Total_cte AS (
        SELECT
        Region,
        District,
        [10_19Days_Total]
    FROM malaria_data
        WHERE Region = 'Lango' AND YEAR = '2020'
        )
    SELECT
        Region,
        SUM([10_19Days_Total]) As total_cases
    FROM Total_cte
GROUP BY Region;


-------------------------------------------------------------------------------------------------------------------

```

```sql

--Total population by  District as per 2021
SELECT
    Year,
    SUM(TotalPopulation) AS total_population
FROM malaria_data
WHERE Year = 2021
  AND LOWER(District) LIKE '% district'
GROUP BY Year;

 --Note: LOWER converts all character in a string to lower Case

-------------------------------------------------------------------------------------------------------------------------------------

--Confirming Total Population by District Level As per 2020 for selected Districts (Ankole Region)
WITH valid_districts AS (
    SELECT *
    FROM malaria_data
    WHERE LOWER(District) LIKE '%district%'
 )
SELECT
        Year,
        SUM(TotalPopulation) AS total_population
   FROM valid_districts
   WHERE Year = 2020
                AND District IN ('Buhweju District','Bushenyi District', 'Ibanda District', 'Isingiro District','Kazo District','Kiruhura District',
    'Mbarara District', 'Mitooma District', 'Ntugamo District', 'Rubirirzi District', 'Rwampara District',
    'Sheema District', 'Mbarara City')
GROUP BY Year;


-------------------------------------------------------------------------------------------------------------------------------------
--Total population Per Region Level(Ankole Region)
SELECT
    Year,
    SUM(TotalPopulation) AS Total_Population
    FROM malaria_data
 WHERE Region = District AND  Year = 2020
      AND Region = 'Ankole'
```

```sql

GROUP BY  Year;

-------------------------------------------------------------------------------------------------------------------------------

--Confiming Total Population For Mbarara District as per 2020
    WITH valid_districts AS (
    SELECT *
    FROM malaria_data
   WHERE LOWER(District) LIKE '%district%'
)

SELECT
    Year,
    SUM(TotalPopulation) AS total_population
FROM valid_districts
WHERE Year = 2020
AND District IN ( 'Mbarara district')
GROUP BY Year;

-------------------------------------------------------------------------------------------------------------------------------


--Find Out How many Districts and Regions  we have in our Dataset As per 2020

/**Districts**/
SELECT COUNT(*) As TotalDistricts
 FRom malaria_data
 WHERE LOWER(District) LIKE '%district%' AND Year = 2020;



 /**Regions**/
--Find Out How many Regions we have
 SELECT COUNT(*) As TotalRegions
 FRom malaria_data
 WHERE District = Region AND Year = 2020;



 ---19/01/2026
 --Retrieving data from an Corrected  Merged  Malaria dataset
 SELECT * FROM malaria_data1;

```

```sql
--------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------------------------

 --U5(<5yr) Cases per Year Strictly(From 2020 to 2024)

WITH Total_cases_cte AS(
    SELECT
           Year,
          SUM( TOTAlCasesConfirmed_BSRDT ) As Total_Cases,
           SUM(
           [0_28Days(BSRDT)_Total]+
           [29Days_4Yrs(BSRDT)_Total]
           ) AS U5TotalCases
    FROM malaria_data1
       WHERE Region = District AND Region <> '0'
    GROUP BY  Year
    )
 SELECT
     Year,
     Total_Cases,
     U5TotalCases,
     CAST(ROUND((CAST((U5TotalCases) AS DECIMAL(18,2)) / CAST
       ((Total_Cases) AS DECIMAL (18,2) )) * 100,2) AS DECIMAL(18,2)) AS
       U5YrBurden
  FROM Total_cases_cte
      ORDER BY Year ASC;

--------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------------------------------

----Percentage Cases Treated Vs Cases Confirmed

WITH treated_cases_cte AS (
    SELECT
           Year,
           SUM(TOTAlCasesConfirmed_BSRDT) As Total_Cases,
           SUM([0_28Days_CT_Total] +
           [29Days_4Yrs_CT_Total] +
           [5_9YRS_CT_Total] +
           [10_19YRS_CT_Total] +
           [20+YRS_CT_Total]) As total_Cases_Treated
        FROM malaria_data1
```

```sql
            WHERE Region = District AND Region <> '0'
         GROUP BY  Year

        )
 SELECT
            Year,
            Total_Cases,
            total_Cases_Treated,
            CAST(ROUND( (CAST((total_Cases_Treated) AS DECIMAL(18,2))/ CAST ((Total_Cases) AS Decimal(18,2)) ) * 100, 2) AS DECIMAL (18,2)) As Percentage_Treatment_Coverage
FROM treated_cases_cte
ORDER BY  Year;


------------------------------------------------------------------------------------------------------------------------

/**Extracting Cases Treated Vs Cases Confirmed**/

WITH treated_cases_cte AS (
        SELECT
               Region,
               Year,
               TOTAlCasesConfirmed_BSRDT As Total_Cases,
               SUM([0_28Days_CT_Total] +
               [29Days_4Yrs_CT_Total] +
               [5_9YRS_CT_Total] +
               [10_19YRS_CT_Total] +
               [20+YRS_CT_Total]) As total_Cases_Treated
            FROM malaria_data1
               WHERE Region = District AND Region <> '0'
            GROUP BY Region,TOTAlCasesConfirmed_BSRDT, Year
           )
 SELECT
       Region,
       Year,
       Total_Cases,
       total_Cases_Treated,
       CAST(ROUND( (CAST((total_Cases_Treated) AS DECIMAL(18,2))/ CAST ((Total_Cases) AS Decimal(18,2)) ) * 100, 2) AS DECIMAL(18,2)) As Percentage_Treatment_Coverage
FROM treated_cases_cte
        ORDER BY region , Year;

------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------

```

```sql

--5-9YRS AGE CATEGORY
/**Extracting Percentage Burden in 5-9yRs Age Group  Per Year Per Region (5-9YRS)**/

--Percentage by Region by Age_group(5_9yrs) by Year
WITH total_Cases_cte AS(
        SELECT
        Region,
        Year,
        TOTAlCasesConfirmed_BSRDT As total_cases, SUM([5_9YRS(BSRDT)_Total]) AS [total_cases_btn(5_9)]
    FROM malaria_data1
        WHERE Region = District AND Region <> '0'
        GROUP BY Region , Year, TOTAlCasesConfirmed_BSRDT
)
    SELECT
         Region,
         Year,
         total_cases, [total_cases_btn(5_9)] ,
         CAST( ROUND((CAST([total_cases_btn(5_9)] AS DECIMAL(18,2)) / CAST( total_cases AS DECIMAL(18,2)) * 100),2) AS DECIMAL(18,2)) AS [5_9_yrs_by_percentage]
    FROM total_Cases_cte ;

---------------------------------------------------------------------------------------------------------------------------------------

----Percentage  By Year(Strictly) per 5_9_Yrs_By_Percentage
    WITH total_Cases_cte AS(
        SELECT
        Year,
        SUM(TOTAlCasesConfirmed_BSRDT) As total_cases, SUM([5_9YRS(BSRDT)_Total]) AS [total_cases_btn(5_9)]
    FROM malaria_data1
        WHERE Region = District AND Region <> '0'
        GROUP BY Year
)
    SELECT

         Year,
         total_cases, [total_cases_btn(5_9)] ,
         CAST( ROUND((CAST([total_cases_btn(5_9)] AS DECIMAL(18,2)) / CAST( total_cases AS DECIMAL(18,2)) * 100),2) AS DECIMAL(18,2)) AS [5_9_yrs_by_percentage]
    FROM total_Cases_cte ;


```

```sql
-------------------------------------------------------------------------------------------------------------------------------
--11/01/2025
-------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------
--ADULTS (10-20+YRS)
/**Extracting Percentage Burden in Adults Per Year Per Region*/
--(10-19yrs) & (20+Yrs) Age Bracket
WITH adult_cte AS(
        SELECT
                   Region ,
                    Year,
                    TOTAlCasesConfirmed_BSRDT AS total_Cases,
                    SUM([10_19YRS(BSRDT)_Total] + [20+YRS_(BSRDT)_Total]) AS Total_Adult_Cases
            FROM malaria_data1
            WHERE Region = District AND Region <> '0'
            GROUP BY Region, Year, TOTAlCasesConfirmed_BSRDT
        )

    SELECT
           Region,
           Year,
           total_cases,
           Total_Adult_Cases,
           CAST(ROUND(((CAST((Total_Adult_Cases) AS DECIMAL(18,2)) / CAST((total_cases) AS DECIMAl(18,2))) * 100),2) AS DECIMAL(18,2)) AS Adult_percentage
    FROM adult_cte ;

-------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------------------
-------------------------------------------------


-----Strictly Percentage by Age Group (ADULT) Per Year

WITH adult_cte AS(
    SELECT
         Year,
         SUM(TOTAlCasesConfirmed_BSRDT) AS total_Cases,
        SUM([10_19YRS(BSRDT)_Total] + [20+YRS_(BSRDT)_Total]) AS Total_Adult_Cases
```

```sql
        FROM malaria_data
            WHERE Region = District AND Region <> '0'
            GROUP BY  Year
            )
    SELECT
           Year,
           total_cases,
           Total_Adult_Cases,
           CAST(ROUND(((CAST((Total_Adult_Cases) AS DECIMAL(18,2)) / CAST
             ((total_cases) AS DECIMAl(18,2))) * 100),2) AS DECIMAL(18,
             2)) AS Adult_percentage
    FROM adult_cte ;
-------------------------------------------------------------------------
-------------------------------------------------------------------------
--------------------------------------
-------------------------------------------------------------------------
-------------------------------------------------------------------------
--------------------------------------
---PERCENTAGE BY GENDER
----FEMALE & MALE

    WITH total_Cases_cte AS(
        SELECT
        Year,
        SUM(TOTAlCasesConfirmed_BSRDT) As total_cases,
        SUM([0_28Days(BSRDT)_Female] +
             [29Days_4Yrs(BSRDT)_Female]+
             [5_9YRS(BSRDT)_Female]+
             [10_19YRS(BSRDT)_Female]+
             [20+YRS_(BSRDT)_Female]) AS Total_female_Cases,
        SUM([0_28Days(BSRDT)_Male] +
             [29Days_4Yrs(BSRDT)_Male]+
             [5_9YRS(BSRDT)_Male]+
             [10_19YRS(BSRDT)_Male]+
             [20+YRS_(BSRDT)_Male]) AS Total_Male_Cases
    FROM malaria_data1
        WHERE Region = District AND Region <> '0'
        GROUP BY Year
)
    SELECT

         Year,
         total_cases, Total_female_Cases, Total_Male_Cases,
         CAST( ROUND((CAST([Total_female_Cases] AS DECIMAL(18,2)) / CAST
           ( total_cases AS DECIMAL(18,2)) * 100),2) AS DECIMAL(18,2)) AS
```

```sql
              Percentage_Total_Cases_Female,
          CAST( ROUND((CAST([Total_Male_Cases] AS DECIMAL(18,2)) / CAST
              ( total_cases AS DECIMAL(18,2)) * 100),2) AS DECIMAL(18,2)) AS
              Percentage_Total_Cases_Male
     FROM total_Cases_cte ;

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